

Dell Pro Max with GB10

System Specification — Compute, OS & NVIDIA Software

Dell's OEM build of the NVIDIA GB10 Grace Blackwell platform, an internal sibling of the NVIDIA DGX Spark. Compute is fixed by the GB10 Superchip; the operating system and library stack are NVIDIA's standard DGX environment.

Compute

SoC	NVIDIA GB10 Grace Blackwell Superchip — unified CPU + GPU package (TSMC 3nm-class)
CPU	20-core Arm v9.2 — 10× Cortex-X925 + 10× Cortex-A725 (MediaTek-produced)
GPU	NVIDIA Blackwell architecture — 6,144 CUDA cores, 5th-gen Tensor Cores
AI performance	Up to 1 petaFLOP (1,000 TFLOPS) of sparse FP4 (~1,000 TOPS)
Memory	128 GB LPDDR5X unified, coherent CPU-GPU pool; 256-bit bus; ~273 GB/s
CPU–GPU link	NVLink-C2C — approximately 5× the bandwidth of PCIe Gen5
Networking	Onboard NVIDIA ConnectX-7: 2× QSFP up to 200 GbE, plus a 10 GbE RJ45 port
Storage	Up to 4 TB NVMe (PCIe Gen4), M.2, with SED hardware encryption
Local model capacity	Inference on models up to ~200B parameters; fine-tuning up to ~70B parameters
Form factor	Dell L6 chassis (model FCM1253), 150 × 150 × 51 mm, ~1.31 kg
Power	~240 W via external USB-C PD adapter

Operating System

Distribution	NVIDIA DGX OS 7 (~7.2.x), based on Ubuntu 24.04 LTS
Kernel	Linux 6.11-based NVIDIA Base OS (NVIDIA-patched)
GPU driver	R580 branch (e.g. 580.95.05+, open-source nvidia driver)
Boot	UEFI by default, with USB-based boot support
Updates	Secure UEFI capsule + LVFS firmware; repository-based over-the-air (OTA) system updates
Operation modes	Desktop (Wayland GUI) or headless via SSH
Management	Out-of-band administration over the 10 GbE RJ45 port or Wi-Fi

NVIDIA Software & Libraries (preinstalled)

CUDA Toolkit	13.x (verify on-device with <code>nvcc --version</code>)
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Deep-learning libraries	cuDNN, NCCL, TensorRT / TensorRT-LLM
Data science	RAPIDS
Containers	NVIDIA Container Toolkit / Container Runtime for Docker
Model & container hub	NGC catalog; NVIDIA NIM inference microservices
Frameworks	PyTorch and others, pulled as NGC containers
Ecosystem	Full NVIDIA AI software stack; work is portable to NVIDIA DGX cloud or data-center systems

Clustering Two Nodes

Two GB10 systems can be bonded into a single logical node for models that exceed one unit's memory — pooling 256 GB of unified memory and up to 8 TB of storage, and enabling distributed inference and fine-tuning of models up to roughly 400B parameters.

Topology	Direct port-to-port link between two units — no switch and no transceivers required
Cable	A single 200G QSFP56 passive Direct Attach Copper (DAC) cable, 0.5 m, NVIDIA-approved part (e.g. Q56-200G-CU0-5); always connect matching port numbers on both nodes
NIC	Onboard NVIDIA ConnectX-7 200 GbE SmartNIC on each unit
Transport	Point-to-point RoCE (RDMA over Converged Ethernet) — a low-latency, high-throughput GPU-to-GPU path
Collectives	NCCL (NVIDIA Collective Communications Library) runs all-reduce / all-gather traffic over the RoCE link; MPI coordinates processes on the CPU side
Combined capacity	256 GB unified memory, up to 8 TB storage, models up to ~400B parameters
Management plane	A separate out-of-band network is still needed (10 GbE RJ45 or Wi-Fi); the DAC carries data only
Setup tooling	NVIDIA Sync “Cluster Assistant” automates IP/subnet and NCCL configuration
Scaling limit	NVIDIA officially supports stacking up to 2 units over a single DAC; 3+ nodes require a 200/400 GbE switch

Version numbers (DGX OS, GPU driver, CUDA, NCCL) reflect the shipping baseline and advance via over-the-air and NGC updates — confirm on the unit with `nvidia-smi` and `nvcc --version`. The Dell Pro Max with GB10 shares the GB10 reference platform with the NVIDIA DGX Spark. Compiled from public NVIDIA, Dell, and independent reviewer documentation, June 2026.